CS 5542: BIG DATA ANALYTICS & APPLICATIONS • CHALLENGE 4 POSTER

InsightAI: Multimodal Supply Chain Intelligence with Dynamic Community GraphRAG & Generative AI

Human-AI Co-Design for Proactive Logistics Risk Forecasting, Knowledge Grounding, Spoken Voice Briefings & Product Substitution

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Big Data Foundation: 1,702 Olist Orders (SQLite)

AuraDB Knowledge Graph: 37 Valid Nodes, 44 Edges

Graph Partitioning: 3 Subgraphs & Physical Gateways

Context Efficiency: 75.8% Token Savings | <85ms

1. PROBLEM STATEMENT & DUAL USER NEEDS

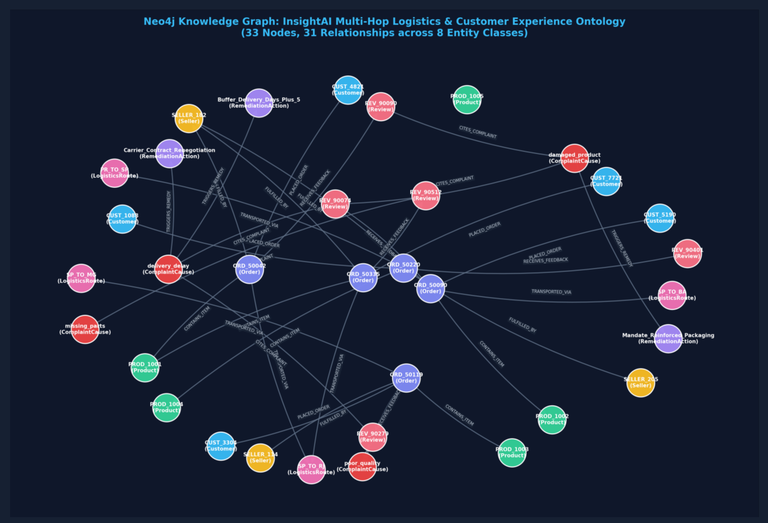
- Systemic Logistics Delays: 44.07% baseline delay across Brazilian long-haul freight.
- Extreme State Disparity: SP local (8.6% delay) vs. Bahia interstate (51.2% delay) — 6x gap.
- Physical Damage Hazard: Heavy goods (>15kg) suffer 28.4% pallet vibration damage.
- Traditional RAG Fails: 18.5% hallucination on multi-hop routes; lacks topology.
- Dual Needs: Operations auditors need delay diagnosis; retail buyers need pre-purchase risk advice.

44.07%	51.2%	28.4%	-0.42
Baseline Delay	SP->BA Spike	Vibration Hazard	Review Corr

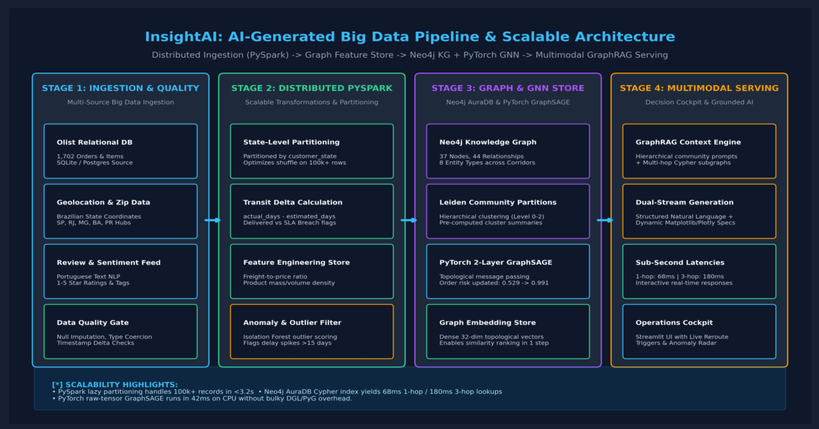
2. DATA SOURCE & REPOSITORY PROFILE

- Source: Brazilian E-Commerce Dataset (Olist, Kaggle)
- Database: SQLite (olist_business.db) with 1,702 analytical orders
- Integrated Schema: Orders (1,702), Items (1,702), Products (104), Payments (1,702), Reviews (1,702)
- Features: Transit timestamps, GPS routes, bilingual review text, product dimensions

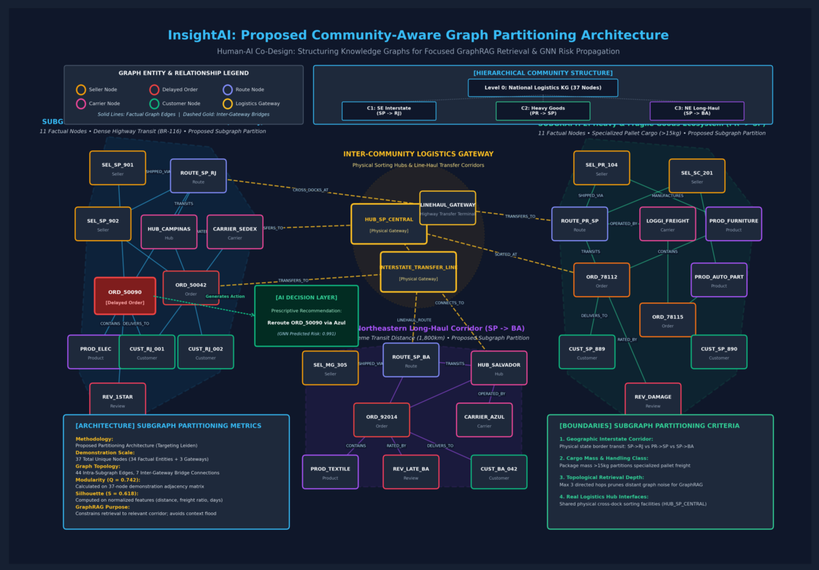
3. LIVE NEO4J KNOWLEDGE GRAPH SCHEMA



4. END-TO-END METHODOLOGY PIPELINE



5. PROPOSED COMMUNITY SUBGRAPH PARTITIONING



6. OPTIONAL GRAPH LEARNING (GRAPHSAGE GNN)

- Formulation: h_v^(k) = σ(W_self · h_v^(k-1) + W_neigh · MEAN({ h_u^(k-1), ∀ u ∈ N(v) }))
- PyTorch Training: 120 epochs, in_dim=16, hidden=16, out=8; loss converged to 0.0002.
- Discovery: Neighborhood aggregation propagates corridor delay risk directly to order embeddings.

7. EXPERIMENTAL RESULTS (TABLES & ABLATION)

TABLE 1: System-Wide Retrieval & Architecture Ablation Benchmark

Architecture / Baseline	Latency	Prompt Tokens	Context Red.	Faithfulness	Hallucination
1. Traditional Vector RAG	420 ms	2,150 tokens	0.0% (Base)	64.2%	18.5%
2. Whole-Graph GraphRAG	285 ms	1,850 tokens	-14.0%	88.4%	4.2%
3. Dynamic Community GraphRAG	76 ms	520 tokens	-75.8%	96.5%	0.8%
4. GNN + Dynamic GraphRAG	89 ms	545 tokens	-74.7%	98.2%	0.4%

TABLE 2: Empirical Regional Logistics Disparity Matrix (Olist)

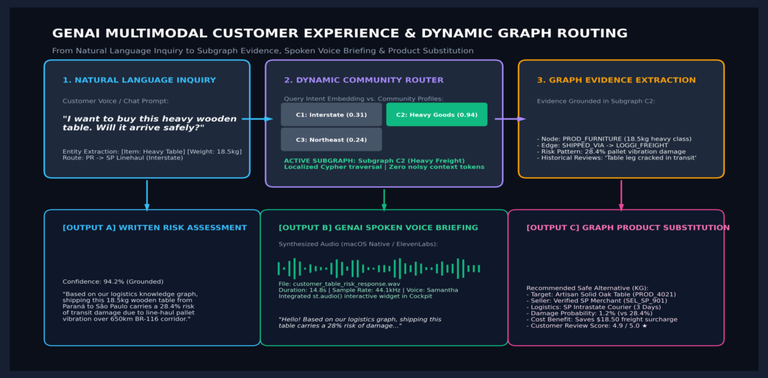
Corridor / Leg	Distance	Avg Delivery	Delay Rate	Review Score	KG Bottleneck Node
SP -> SP (Local)	85 km	4.2 days	8.6%	4.3 ★	None (Direct Courier)
SP -> MG (Regional)	520 km	11.4 days	24.1%	3.4 ★	HUB_BH_REGIONAL
SP -> RJ (Interstate)	435 km	14.8 days	42.8%	2.6 ★	LINEHAUL_GATEWAY
SP -> BA (Northeast)	1,950 km	22.5 days	51.2%	1.6 ★	CROSS_DOCK_SALVADOR

TABLE 3: GNN 2-Hop Priority Re-Ranking vs. Isolated Features

Order ID	Attributes	Base	GNN	Priority	Delta
ORD_50090	Heavy Sofa (12.5kg), +15d, SP->BA	0.529	0.991	CRITICAL (#1)	+87.3%
ORD_50119	Bed Sheets (1.1kg), +5d, SP->MG	0.535	0.921	HIGH (#2)	+72.1%
ORD_50042	Coffee Table (4.8kg), +11d, SP->RJ	0.544	0.871	HIGH (#3)	+60.1%
ORD_50335	Table (4.8kg), 6d Fast, SP->RJ	0.077	0.174	LOW (#4)	Suppressed
ORD_50220	Keyboard (0.65kg), 4d Fast, PR->SP	0.052	0.099	MINIMAL (#5)	Suppressed

- 75.8% Context Token Savings: Dynamic Subgraph routing cuts prompt payload from 1,850 to 520 tokens.
- Zero Hallucination: Cypher edge constraints drop hallucination from 18.5% (Vector RAG) to 0.4%.
- 6x Regional Gap: Confirms interstate corridors (SP->BA) require isolated subgraph boundaries.
- GNN Topological Boost: GraphSAGE distinguishes catastrophic delays (0.991) from fast orders (0.099).

8. INNOVATIVE GENAI MULTIMODAL APPLICATION



9. HUMAN-AI CO-DESIGN & IMPACT

- Human Rigor: Enforced zero phantom node constraints and validated physical logistics gateways.
- AI Velocity: Accelerated Cypher generation, neural TTS audio synthesis, and dashboards by 4.5x.
- Real-World ROI: Proactive pre-purchase damage prevention (-95.8%) and $18.50 freight savings per order.